

Student Handbook

Docker Training

Day 1 · Session 2

Writing Dockerfiles for .NET Web API & Angular

10:45 – 12:15 · 1.5 hours · Hands-on Lab

Prerequisites: Completed Day 1 Session 1 — Docker is installed and running

Learning objectives

By the end of this session you will be able to:

- Explain every Dockerfile instruction (FROM, WORKDIR, COPY, RUN, ENV, EXPOSE, ENTRYPOINT, CMD)
- Write a single-stage Dockerfile and understand its limitations
- Write a multi-stage Dockerfile for .NET 8 Web API with a dramatically smaller output image
- Write a multi-stage Dockerfile for Angular using Node to build and Nginx to serve
- Use .dockerignore to exclude build artefacts and speed up builds
- Understand how layer caching works and how to order instructions to exploit it
- Compare Alpine vs Debian base images and choose the right one for your use case

Dockerfile instructions — reference

A Dockerfile is a plain text file with one instruction per line. Docker reads it top to bottom, executing each instruction as a new read-only layer on top of the previous one.

Instruction	What it does
FROM <image>:<tag>	Sets the base image for all subsequent instructions. Must be the first instruction.
WORKDIR /path	Sets (and creates if needed) the working directory for RUN, COPY, ENTRYPOINT, CMD.
COPY <src> <dest>	Copies files from the build context (your machine) into the image.
RUN <command>	Executes a shell command at build time and commits the result as a new layer.
ENV KEY=value	Sets an environment variable visible at both build time and runtime.
ARG NAME=default	Build-time variable only. Not available in the final image or at runtime.
EXPOSE <port>	Documents which port the container listens on. Does NOT actually publish the port.
ENTRYPOINT ["cmd"]	The executable to run when the container starts. Hard to override.
CMD ["arg"]	Default arguments passed to ENTRYPOINT. Easily overridden with docker run <image> <cmd>.
USER <name>	Switches to a non-root user for all subsequent RUN / CMD / ENTRYPOINT instructions.
.dockerignore	Not an instruction — a file in the build context that excludes paths from COPY.

How layer caching works

Docker caches the result of each instruction. On a rebuild, if an instruction and everything above it are unchanged, Docker reuses the cached layer — making rebuilds very fast.

The golden rule of caching

Copy things that change LAST. If you copy your source code before installing dependencies, every code change busts the dependency cache and forces a full npm install or dotnet restore.

Cache-friendly order for a .NET project:

- COPY *.csproj first — project file changes rarely
- RUN dotnet restore — cached as long as .csproj hasn't changed
- COPY . . — copy all source code
- RUN dotnet publish — runs only when source actually changed

Cache-friendly order for an Angular project:

- COPY package.json package-lock.json first — lockfile changes rarely
- RUN npm ci — cached as long as lockfile hasn't changed
- COPY . . — copy all source
- RUN ng build — runs only when source changed

Multi-stage builds

A multi-stage build uses multiple FROM instructions in one Dockerfile. Each FROM starts a fresh stage. You selectively COPY artefacts from one stage into the next — leaving behind compilers, SDKs, node_modules, and anything not needed at runtime.

Single-stage .NET image

Includes the full .NET SDK (compilers, MSBuild, templates). Final image ≈ 850 MB — shipped to production!

Multi-stage .NET image

Stage 1 uses sdk:8.0 to compile. Stage 2 uses aspnet:8.0 (runtime only). Final image ≈ 220 MB — only the app and runtime.

Hands-on lab steps

Create the .NET 8 Web API project

1

Scaffold a minimal project you will containerise throughout this session

Create a new Web API project

```
# Create a working directory for today's labs
mkdir docker-training && cd docker-training

# Scaffold a new .NET 8 Web API
dotnet new webapi -n WeatherApi --no-openapi
cd WeatherApi

# Run it locally first to confirm it works
dotnet run
# Visit https://localhost:5001/weatherforecast in your browser
# Press Ctrl+C to stop
```

Explore the project structure

```
ls -la
# You should see: WeatherApi.csproj Program.cs Controllers/
```

Note

The goal is to containerise this exact project. You are not changing any application code in this session — only the Docker configuration around it.

2

Write a single-stage Dockerfile for .NET (and see the problem)

Understand why single-stage is not suitable for production

Create Dockerfile.single in the WeatherApi folder

```
# Create the file
touch Dockerfile.single

# Open it in your editor and paste the content below
```

Dockerfile.single — content

```
FROM mcr.microsoft.com/dotnet/sdk:8.0
WORKDIR /app

# Copy project file first (cache the restore step)
COPY WeatherApi.csproj .
RUN dotnet restore

# Copy all source code
COPY . .

# Build and publish
RUN dotnet publish -c Release -o /app/publish

EXPOSE 8080
ENTRYPOINT ["dotnet", "/app/publish/WeatherApi.dll"]
```

Build and check the image size

```
docker build -f Dockerfile.single -t weatherapi:single .

docker images weatherapi
```

Expected output:

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
weatherapi	single	alb2c3d4e5f6	8 seconds ago	~850MB

What is wrong here?

The final image contains the entire .NET SDK — compilers, MSBuild, templates, NuGet cache. None of that is needed to run the app. You are shipping 630 MB of build tooling to every production server.

3 Write a multi-stage Dockerfile for .NET (production-ready)

Stage 1 compiles with the SDK. Stage 2 runs with the lightweight runtime.

Create Dockerfile in the WeatherApi folder

```
touch Dockerfile
# Open it in your editor and paste the content below
```

Dockerfile — full content

```
# — Stage 1: Build —————
FROM mcr.microsoft.com/dotnet/sdk:8.0 AS build
WORKDIR /src

# Copy project file first so dotnet restore is cached separately
COPY WeatherApi.csproj .
RUN dotnet restore

# Copy remaining source and publish
COPY . .
RUN dotnet publish -c Release -o /app/publish --no-restore

# — Stage 2: Runtime —————
FROM mcr.microsoft.com/dotnet/aspnet:8.0 AS runtime
WORKDIR /app

# Copy ONLY the published output from the build stage
COPY --from=build /app/publish .

# Run as non-root for security (covered in depth on Day 4)
RUN adduser --disabled-password --gecos '' appuser && chown -R appuser /app
USER appuser

EXPOSE 8080
ENTRYPOINT ["dotnet", "WeatherApi.dll"]
```

Build the multi-stage image

```
docker build -t weatherapi:1.0 .
```

```
# Compare sizes side by side
docker images weatherapi
```

Expected output:

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
weatherapi	1.0	b2c3d4e5f6a7	12 seconds ago	~220MB
weatherapi	single	alb2c3d4e5f6	5 minutes ago	~850MB

Run and test the multi-stage image

```
docker run -d --name api -p 8080:8080 weatherapi:1.0

# Test the endpoint
curl http://localhost:8080/weatherforecast

# Or open http://localhost:8080/weatherforecast in your browser
```

Result

The production image is ~220 MB — 630 MB smaller — and contains only the ASP.NET runtime and your compiled app. No SDK. No source code. No build tools.

4

Create .dockerignore — speed up builds and reduce context size

Tell Docker what NOT to send to the daemon during builds

Every time you run `docker build`, Docker sends the entire build context (the folder you specify) to the daemon. Without `.dockerignore`, this includes `bin/`, `obj/`, and `.git/` — often hundreds of megabytes of files that are never used.

Create .dockerignore in the WeatherApi folder

```
# Create the file
touch .dockerignore
# Open in your editor and paste:
```

.dockerignore — content

```
# Build output — always generated fresh inside the container
bin/
obj/

# Git history — never needed in an image
.git/
.gitignore

# Docker files themselves
Dockerfile*
.dockerignore

# Editor and OS files
.vs/
.vscode/
*.user
.DS_Store
```

```
# Test projects (don't ship tests to production)
**/*.Tests/
**/*.Test/
```

Verify the context size difference

```
# Remove old image, rebuild clean – watch the 'Sending build context' line
docker rmi weatherapi:1.0
docker build -t weatherapi:1.0 .
```

Expected output:

```
# Without .dockerignore you might see:
=> [internal] load build context   2.3s
=> => transferring context: 85.2MB

# With .dockerignore:
=> [internal] load build context   0.1s
=> => transferring context: 12.3kB
```

Why this matters

A smaller build context means faster builds, especially on CI servers where the context is uploaded over a network. It also prevents accidentally leaking secrets or credentials that live in your project folder.

5 Write a multi-stage Dockerfile for Angular

Node builds the app. Nginx serves the static files. The result is tiny.

Create the Angular project (from the docker-training root)

```
cd ..    # back to docker-training/

# Create a new Angular app (skip git init)
npx @angular/cli@latest new weather-ui --routing=false --style=css --skip-git
cd weather-ui
```

Create Dockerfile in the weather-ui folder

```
touch Dockerfile
# Paste the content below
```

Dockerfile — full content

```
# — Stage 1: Build —————
FROM node:20-alpine AS build
WORKDIR /app

# Copy lockfiles first – npm ci is cached until these change
COPY package.json package-lock.json .
RUN npm ci

# Copy source and build for production
COPY . .
RUN npx ng build --configuration production

# — Stage 2: Serve —————
```

```
FROM nginx:1.27-alpine AS runtime

# Remove default Nginx page
RUN rm -rf /usr/share/nginx/html/*

# Copy Angular build output
COPY --from=build /app/dist/weather-ui/browser /usr/share/nginx/html

# Optional: custom nginx config (see tip below)
# COPY nginx.conf /etc/nginx/conf.d/default.conf

EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

Create .dockerignore for Angular

```
# .dockerignore
node_modules/
dist/
.git/
.gitignore
Dockerfile*
.dockerignore
.angular/
.vscode/
*.md
```

Build and run the Angular image

```
docker build -t weather-ui:1.0 .
```

```
# Compare sizes
docker images weather-ui
```

Expected output:

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
weather-ui	1.0	c3d4e5f6a7b8	15 seconds ago	~45MB

```
# Run the UI on port 4200
docker run -d --name ui -p 4200:80 weather-ui:1.0
```

```
# Open http://localhost:4200 — you should see the Angular welcome page
```

What just happened

node_modules (~300 MB) and the Angular compiler were used in Stage 1 and then discarded. The final image is Nginx + your compiled JS/HTML/CSS — only ~45 MB.

6

Demonstrate layer caching in action

Change only the source code and watch what gets re-used

This step shows caching in a concrete, measurable way.

First build — everything runs from scratch


```
# In weather-ui/ - time the first build
time docker build -t weather-ui:cache-test .
```

Expected output:

```
=> [build 2/5] COPY package.json package-lock.json .      0.1s
=> [build 3/5] RUN npm ci                                   42.3s ← slow
=> [build 4/5] COPY . .                                     0.2s
=> [build 5/5] RUN npx ng build --configuration prod       18.1s ← slow

real    1m2.4s
```

Change only a source file — then rebuild

```
# Simulate a source code change
echo "<!-- updated -->" >> src/app/app.component.html
```

```
# Rebuild
time docker build -t weather-ui:cache-test .
```

Expected output:

```
=> CACHED [build 2/5] COPY package.json package-lock.json . 0.0s
=> CACHED [build 3/5] RUN npm ci                             0.0s ← cached!
=> [build 4/5] COPY . .                                     0.2s
=> [build 5/5] RUN npx ng build --configuration prod       17.8s

real    0m18.3s
```

Result

42 seconds saved on npm ci because the package files didn't change. In a real project with hundreds of packages, this is the difference between a 3-minute CI build and a 30-second one.

Alpine vs Debian base images

Factor	Alpine (alpine, -alpine)	Debian (bookworm-slim, etc.)
Base image size	~5 MB	~70–100 MB
Package manager	apk	apt-get
Shell available	sh (not bash by default)	bash
.NET compatibility	Use mcr.microsoft.com/dotnet/aspnet:8.0-alpine	Use aspnet:8.0 (default)
Node compatibility	node:20-alpine Works well	node:20 (larger)
Common issues	Missing glibc — some native npm packages fail	Very compatible — rarely breaks
Best for	Node/Angular, simple Go/Python apps	.NET, apps with native dependencies

Quick reference

Command	What it does
<code>docker build -t name:tag .</code>	Build image from Dockerfile in current directory
<code>docker build -f Dockerfile.dev -t name:tag .</code>	Build using a specific Dockerfile name
<code>docker build --no-cache -t name:tag .</code>	Force rebuild — ignore all cached layers
<code>docker images</code>	List all local images with sizes
<code>docker rmi name:tag</code>	Remove a local image
<code>docker image prune</code>	Remove all dangling (untagged) images
<code>docker history name:tag</code>	Show all layers and their sizes
<code>docker run -d --name x -p HOST:CONT img</code>	Run container in background with port map
<code>docker rm -f name</code>	Force stop and remove a container

Common mistakes to avoid

Mistake 1 — Copying source before restoring packages

`COPY . .` before `RUN dotnet restore` or `RUN npm ci` means every code change forces a full package restore. Always copy only the package manifest first, restore, then copy source.

Mistake 2 — No `.dockerignore`

Without `.dockerignore`, the build context includes `node_modules/` (hundreds of MB), `bin/`, `obj/`, and `.git/` — all sent to the Docker daemon on every build. Always add `.dockerignore` before your first build.

Mistake 3 — Using the tag 'latest'

FROM `node:latest` changes silently when the upstream releases a new version. Pin to a specific tag like `node:20-alpine` to make builds reproducible.

Mistake 4 — Forgetting `--from=build` in multi-stage `COPY`

`COPY /app/publish .` copies from the build context (your machine). `COPY --from=build /app/publish .` copies from the earlier build stage. Always include `--from=` in multi-stage Dockerfiles.